

Alan Zheng

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// EXPERIENCE

SDE II @ Amazon Web Services

Aug 2023 – Present

IAM Data Plane • Seattle, WA

- ▶ Led migration of authentication traffic across 11 global regions to a sharded architecture with zero customer impact.
- ▶ Designed automated rollback for migrating production traffic to a new auth stack, using the legacy stack as a live statistical control (z-tests, anomaly detection) and metrics verified with six adopting teams.
- ▶ Built core authentication APIs in Rust reaching full feature parity with legacy Java services, then designed and built client authentication and authorization with the authorization library team.
- ▶ Cut auth-API latency ~45% through systematic profiling and allocation, I/O, and framework-overhead optimization; stood up isolated load-test environments that gate releases on performance regressions.
- ▶ Root-caused a production tail-latency step-up on a tight launch timeline, tracing it to a lock-contended per-request lookup; caching it at startup made the affected path ~96% faster.
- ▶ Root-caused a Rust production memory leak via heap-profile diffs; growth fell from 1 GB/27 min to 0.4 MB/80 min.
- ▶ Unblocked a month-stalled production launch by standing up mutual-TLS service-to-service communication in one week, migrating the transport layer to a Netty-based stack with automated certificate rotation.
- ▶ Built a security scanner daemon that proactively detects data-consistency vulnerabilities, surfacing 50+ issues; now a mandatory release gate.
- ▶ Mentored two interns (both received return offers) and onboarded senior engineers to the auth domain.

SDE Intern @ Amazon Web Services

Summers 2021 & 2022

IAM Data Plane • Seattle, WA

- ▶ Built a CLI debugging tool for on-call engineers to diagnose IAM policy evaluation issues in production.
- ▶ Designed a DynamoDB-backed bootstrap rate limiter with metrics and deployment scripts to cut maintenance load during fleet scaling.

Teaching Assistant & Tutor @ UVA Engineering

Sep 2021 – May 2023

Charlottesville, VA

- ▶ TA for cryptocurrency technologies; tutored linear algebra, vector calculus, machine learning, and data structures.

// PROJECTS

InterviewCoach

Python • LangGraph • AWS Bedrock • Multi-Agent

Agentic mock-interview app: a LangGraph orchestrator runs the interview while a parallel evaluator swarm scores each answer per dimension. Built on AWS Bedrock (Claude Sonnet + Haiku) with a swappable vLLM backend and a Streamlit UI.

Multiplayer Tetris with RL

Python • PyTorch • Deep Q-Learning • MARL

Trained multi-agent RL agents with Deep Q-Learning, shared parameters, and joint state modeling in a custom multiplayer Tetris environment. Achieved 87% win rate vs. a single-agent baseline.

Rust Ops CLI

Rust • AWS S3 • CLI • Ops Tooling

Rust CLI for pipeline health monitoring, performance benchmarking with S3 storage and run comparison, host scanning, and deployment diagnostics. Used daily by the team.

Drone Swarm Research

Python • RL • ROS • Vicon

RL model controlling a swarm of rescue drones in unknown environments. Produced a research paper and live demo.

// SKILLS

Languages

Rust, Java, Python, JavaScript, C++, Bash, SQL

AI / ML

PyTorch, TensorFlow, Transformers, scikit-learn, OpenCV, spaCy, NumPy, Pandas

Cloud & Infra

AWS (IAM, Lambda, EC2, S3, DynamoDB, Bedrock), Docker, GitHub Actions, Jenkins

Security & Crypto

TLS / mTLS, SigV4, ECDSA, HMAC, X.509, cipher suites

Performance

CodeGuru, perf, flamegraphs, heap & latency profiling

// EDUCATION

University of Washington • MS, Computer Science & Engineering • GPA 4.0

Expected Jun 2027

University of Virginia • BS, Computer Science • GPA 3.98 • Minors: Applied Math, Data Science

May 2023